

Name: **Suggested Solutions**



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

Class	Reg Number
17	
Calculator Model / No.	

Chemistry

9729/02

Paper 2 Structured Questions

18 September 2018

2 hours

Additional Materials: *Data Booklet*

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

Write your calculator brand and model/number in the box provided above.

Answer **all** questions in the spaces provided on the question paper.

All working must be shown clearly.

INFORMATION FOR CANDIDATES

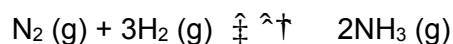
The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

Examiner's Use			
Paper 1	MCQ	/ 30	/ 15 %
Paper 2	Q1	/ 13	
	Q2	/ 19	
	Q3	/ 14	
	Q4	/ 15	
	Q5	/ 14	
	Total	/ 75	/ 30%
Paper 3	Total	/ 80	/ 35%
Paper 4	Total	/ 55	/ 20%
Total	/ 100 %		
Grade			

This document consists of **17** printed pages (including this cover page).

- 1 Ammonia is one of the most highly produced inorganic chemicals. Modern chemical plants depend on Haber Process whereby hydrogen derived from natural gas reacts with atmospheric nitrogen using a catalyst under high pressure and temperature to produce anhydrous ammonia.



- (a) For the above reaction, a 1:3 ratio of N_2 and H_2 is passed over a catalyst. The resulting mixture is allowed to reach dynamic equilibrium. At equilibrium, 20% of the N_2 has reacted and the total pressure is 2.0 atm.

- (i) Explain what is meant by the term *dynamic equilibrium*.

[1]

A dynamic equilibrium is a chemical equilibrium where the rate of forward reaction is equal to the rate of backward reaction and the substances are still reacting together although the concentrations of the reactants and products remain constant.

- (ii) Calculate the partial pressures of N_2 and H_2 at equilibrium.

[2]

	$\text{N}_2(\text{g})$	+	$3\text{H}_2(\text{g})$	$\rightleftharpoons$	$2\text{NH}_3(\text{g})$
Initial no. of mole	x		3x		0
Change in no. of mole	- 0.2 x		- 0.6 x		+ 0.4 x
No. of mole at equilibrium	0.8 x		2.4 x		0.4 x

Total no. of moles at equilibrium = $0.8x + 2.4x + 0.4x = 3.6x$

Mole fraction at equilibrium	$0.8x / 3.6x$ = 0.222	$2.4x / 3.6x$ = 0.667	$0.4x / 3.6x$ = 0.111
Partial Pressure at equilibrium/ atm	0.222×2.0 = 0.444	0.667×2.0 = 1.33	0.111×2.0 = 0.222

Partial pressure of N_2 at equilibrium = 0.444 atm

Partial pressure of H_2 at equilibrium = 1.33 atm

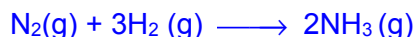
- (iii) Write an expression for the equilibrium constant, K_p , and calculate its value.

[2]

$$K_p = \frac{(P_{\text{NH}_3})^2}{(P_{\text{N}_2})(P_{\text{H}_2})^3} = \frac{(0.222)^2}{(0.444)(1.33)^3} = 0.0472 \text{ atm}^{-2}$$

- (b) (i) Using relevant data in the *Data Booklet*, calculate the enthalpy change of reaction, ΔH_{rxn} of the Haber Process.

[2]



$$\begin{aligned}\Delta H_{\text{rxn}} &= [\text{BE}(\text{N}\equiv\text{N}) + 3\text{BE}(\text{H}-\text{H}) - 6\text{BE}(\text{N}-\text{H})] \\ &= [(944) + 3(436)] - [6(390)] \\ &= -88.0 \text{ kJ mol}^{-1}\end{aligned}$$

- (ii) The enthalpy change of reaction calculated above in (b)(i) differs slightly from the actual value of $-92.0 \text{ kJ mol}^{-1}$. Suggest a reason for this.

[1]

The bond energy calculation is an approximation method as the N–H bond energy value / bond energy values for polyatomic molecules from the *Data Booklet* are average values.

- (c) Explain whether the formation of ammonia is favoured with

- I high or low pressure,
II high or low temperature.

[3]

- I High pressure favours ammonia formation.

By Le Chatelier's Principle, with an increase in pressure, the equilibrium position will shift right to decrease the number of moles of gas to decrease pressure.

New equilibrium mixture contains less reactants, N_2 and H_2 and more products, NH_3 . Yield of ammonia increases.

- II Low temperature favours ammonia formation.

By Le Chatelier's Principle, with a decrease in temperature, the equilibrium position will shift right towards exothermic reaction to release heat.

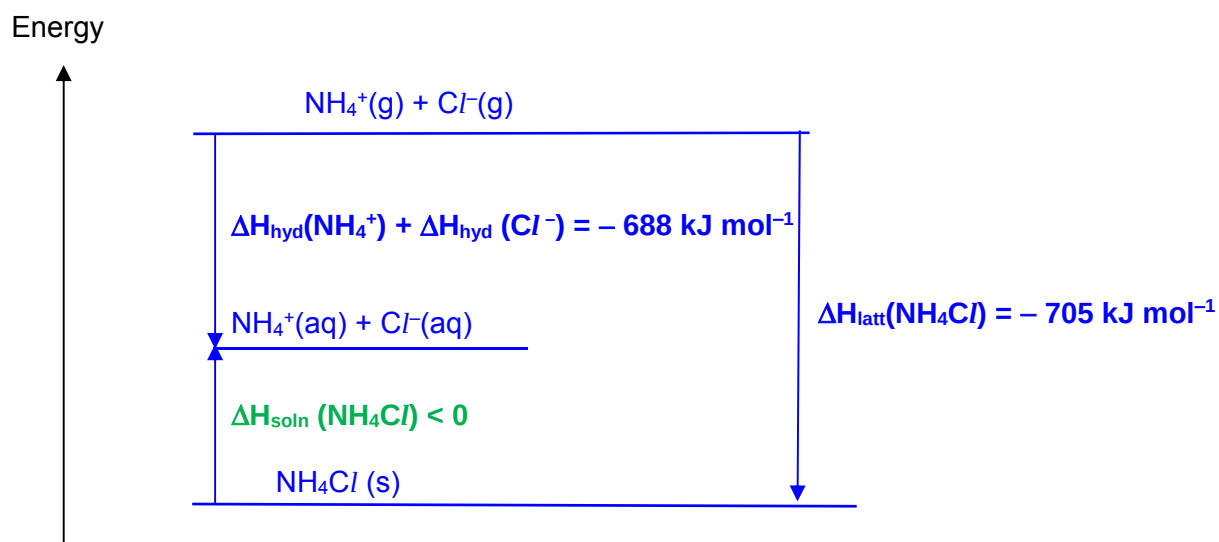
New equilibrium mixture contains less reactants, N_2 and H_2 and more products, NH_3 . Yield of ammonia increases.

- (d) Ammonium chloride, NH_4Cl is typically used in instant ice-packs. A crushing action activates the ice-pack as this mixes NH_4Cl with water. The ice-pack cools as the dissolution of NH_4Cl is endothermic.

Draw an energy level diagram to determine the standard enthalpy change of solution of ammonium chloride given the following information.

Lattice energy of ammonium chloride	-705 kJ mol^{-1}
Enthalpy change of hydration of NH_4^+	-307 kJ mol^{-1}
Enthalpy change of hydration of Cl^-	-381 kJ mol^{-1}

[2]



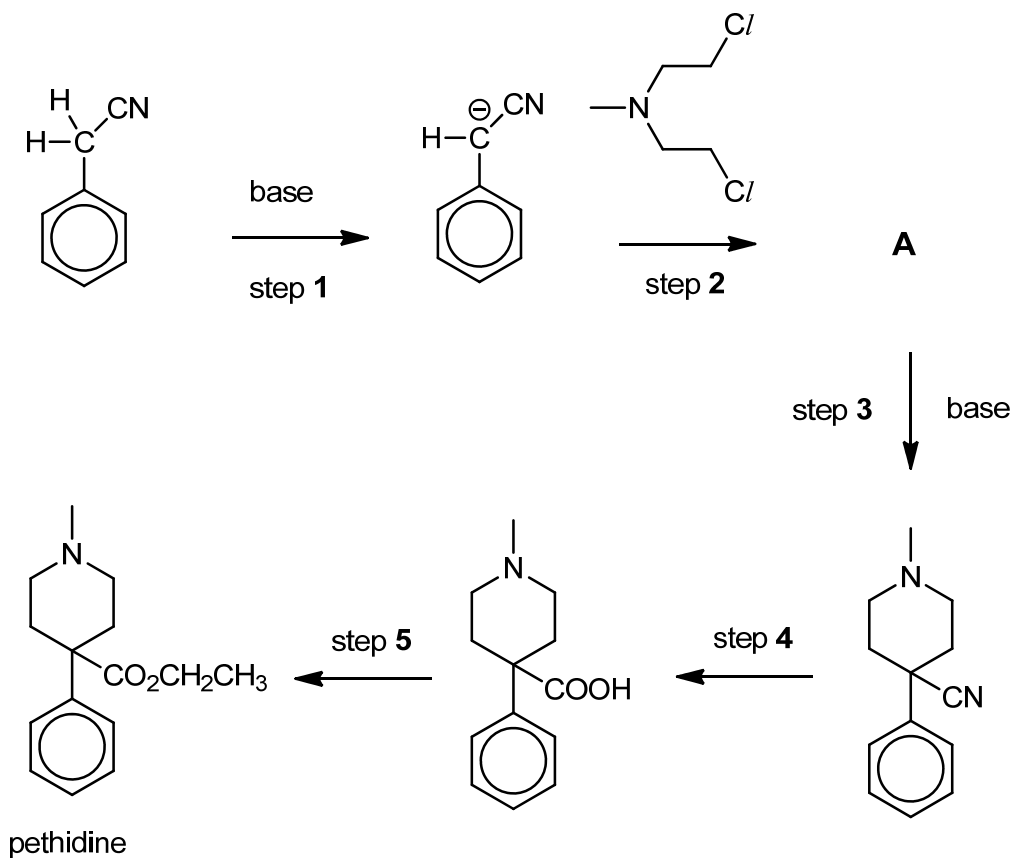
By Hess' Law,

$$\begin{aligned}
 \Delta H_{\text{soln}} (\text{NH}_4\text{Cl}) &= -\Delta H_{\text{latt}} (\text{NH}_4\text{Cl}) + \Delta H_{\text{hyd}}(\text{NH}_4^+) + \Delta H_{\text{hyd}} (\text{Cl}^-) \\
 &= -(-705) + (-307) + (-381) \\
 &= +17.0 \text{ kJ mol}^{-1}
 \end{aligned}$$

[Total: 13]

2 Nitrogen is an element commonly found in various major classes of drugs.

(a) The following is a reaction scheme for the synthesis of pethidine, a powerful painkilling drug.



(i) State the type of reaction for step 1 and step 2.

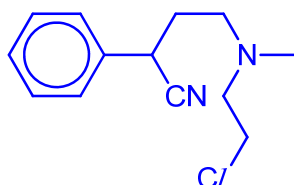
[2]

Step 1: acid–base reaction / neutralisation

Step 2: nucleophilic substitution

(ii) Suggest the structure of intermediate A

[1]



(iii) State the reagents and conditions needed for step 4 and step 5.

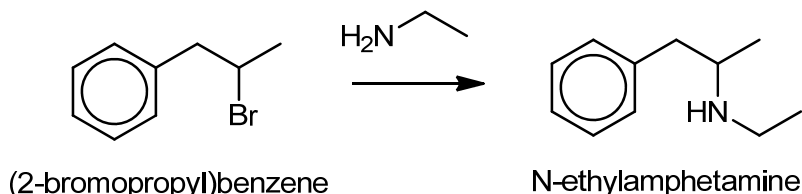
[2]

Step 4: HCl (aq) / H₂SO₄ (aq), heat under reflux

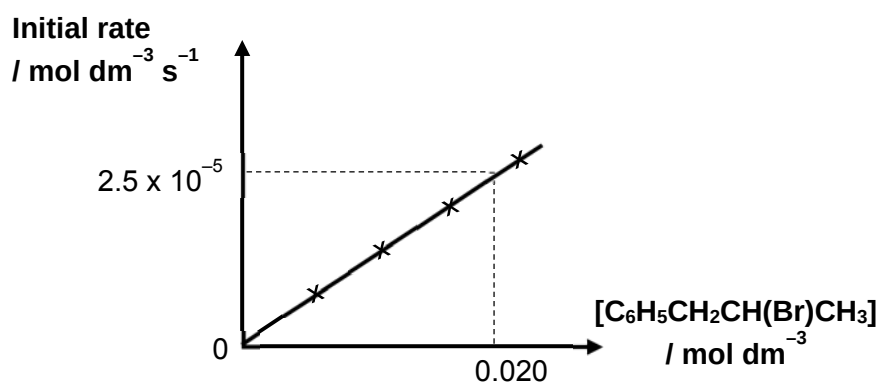
Step 5: CH₃CH₂OH, conc. H₂SO₄, heat OR PCl₅ followed by CH₃CH₂OH

- (b) Kinetics studies of organic reactions provide strong evidence in supporting the postulated mechanisms.

The results from the kinetics study of the nucleophilic substitution reaction involved in the synthesis of N-ethylamphetamine, a discontinued weight-loss drug, are presented below.



The graph was plotted based on data obtained from repeating the experiment several times by varying the concentration of $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$ while maintaining that of $\text{CH}_3\text{CH}_2\text{NH}_2$ to be at 2.5 mol dm^{-3} .



- (i) The rate equation for the reaction was found to be

$$\text{rate} = k [\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3][\text{CH}_3\text{CH}_2\text{NH}_2]$$

Using the graph provided, justify why the order of reaction with respect to $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$ is one.

[2]

The graph of rate against $[\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3]$ is a positively sloped straight line passing through the origin hence the initial rate is directly proportional to $[\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3]$.

- (ii) Calculate the rate constant, k and state its units.

[2]

Using data from the graph,

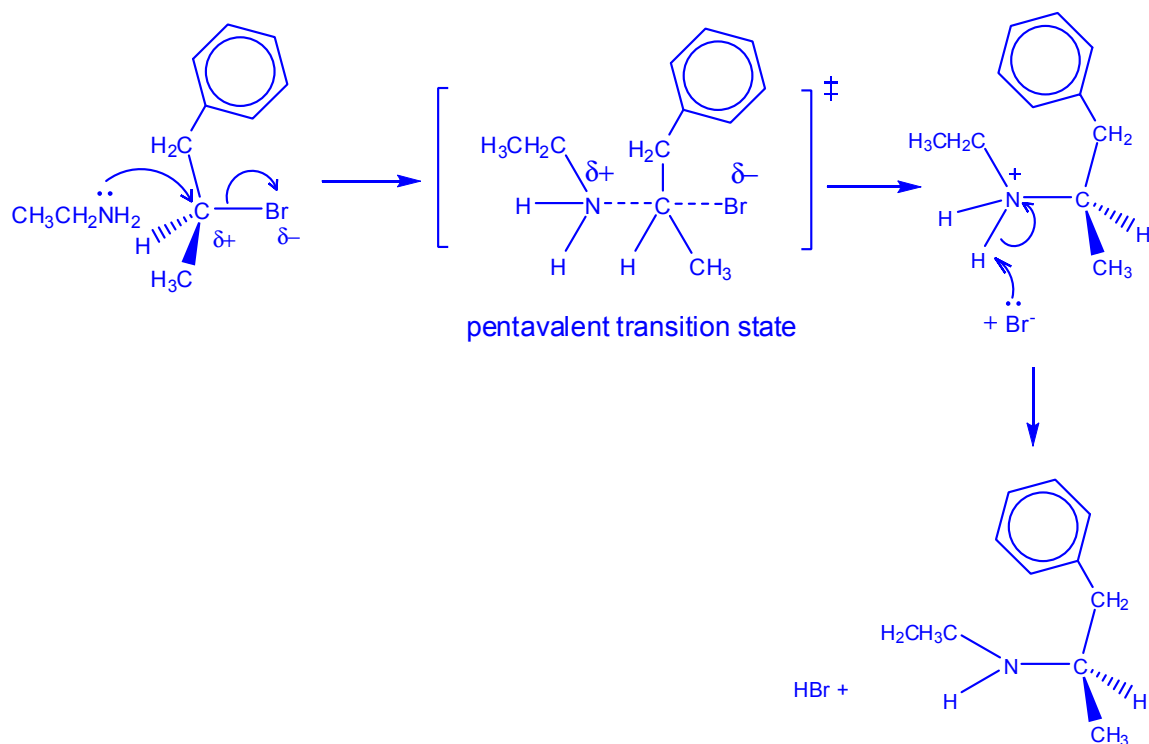
$$\text{Rate} = k [\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3] [\text{CH}_3\text{CH}_2\text{NH}_2]$$

$$2.5 \times 10^{-5} = k (0.020)(2.5)$$

$$k = 5.00 \times 10^{-4} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

- (iii) Given that this is a bimolecular reaction, outline the mechanism for this reaction. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons. [3]

Nucleophilic Substitution (S_N2)



- (iv) Suggest why the rate of reaction decreases when (CH₃)₃CNH₂ is used instead of CH₃CH₂NH₂. [1]

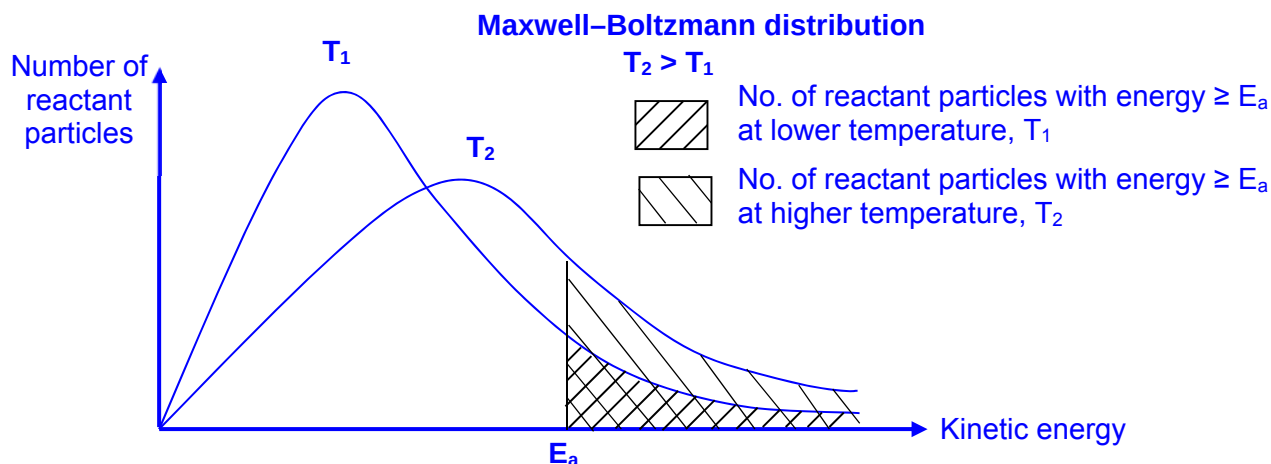
The alkyl group of (CH₃)₃CNH₂ is bulkier than that of CH₃CH₂NH₂ hence there will be steric hindrance. The rate of reaction will decrease.

- (v) Explain how the reactivity of (2-chloropropyl)benzene will differ towards CH₃CH₂NH₂ as compared to (2-bromopropyl)benzene. [1]

C-Cl bond is stronger than C-Br. Hence (2-chloropropyl)benzene will be less reactive towards CH₃CH₂NH₂.

- (vi) With the aid of a Maxwell–Boltzmann Distribution curve, explain how heating the reaction mixture will increase the rate of the reaction.

[3]



- When temperature is increased, number of reactant particles with energy $\geq E_a$ increases,
- Frequency of effective collisions increases.
- Since rate of reaction is proportional to the frequency of effective collisions, rate of reaction increases.

- (c) Investigating the optical activity of the product of a nucleophilic substitution reaction can also provide evidence on how the reaction may proceed.

Consider the reaction between the (+) enantiomer of 2-bromobutane and hot NaOH (aq).

State the expected optical activity of the product if this reaction proceeds via S_N1 mechanism. Provide reasoning for your answer.

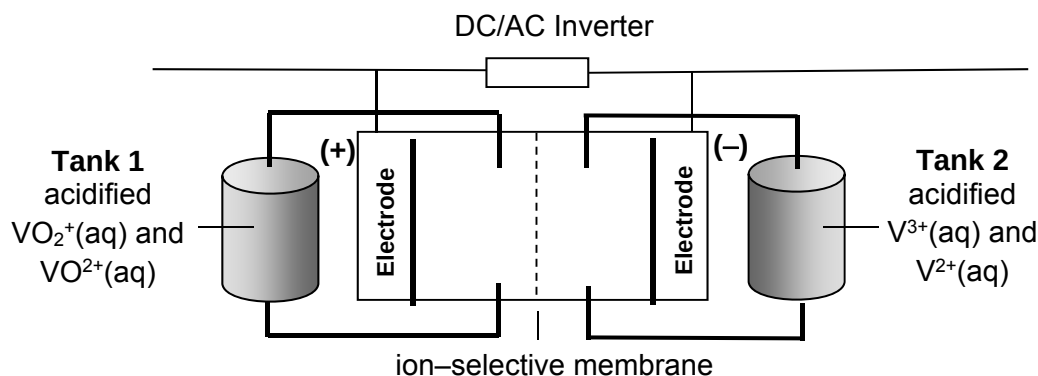
[2]

If the reaction proceeds via a S_N1 mechanism, the product obtained is likely going to be optically inactive.

Since S_N1 mechanism involves the formation of a carbocation intermediate, the nucleophile (OR OH^-) can attack the sp^2 hybridised (OR trigonal planar) carbocation (OR $[\text{CH}_3\text{CH}(\text{CH}_2\text{CH}_3)]^+$) intermediate from either top and bottom of the plane with equal probability giving rise to equal amount of both enantiomers of the product (OR racemic mixture).

[Total: 19]

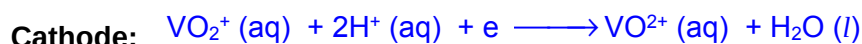
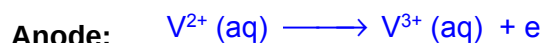
- 3 The vanadium flow battery (VFB) is a type of rechargeable battery that employs electrolyte containing vanadium ions in different oxidation states.



- (a) Electricity is generated from the battery by half-cell reactions of these electrolytes that were pumped into the cell from separate storage tanks. The two electrolytes in the cell are separated by an ion selective membrane, which only allows H^+ ions to pass through.

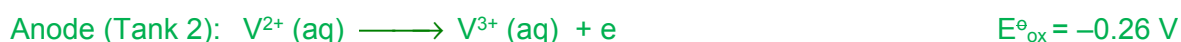
- (i) Write the two half equations for the reaction occurring at the anode and the cathode when the battery discharges. Hence, write the overall cell reaction.

[2]



- (ii) Calculate the e.m.f of the battery.

[1]



$$E^\ominus_{\text{cell}} = E^\ominus_{\text{red}} - E^\ominus_{\text{oxid}} = +1.00 - (-0.26) = +1.26 \text{ V}$$

- (iii) Explain how the ion-selective membrane in VFB maintains electrical neutrality.

[1]

As the reaction proceeds, the anode/Tank 2 becomes more positively charged (formation of more positively charged ion) whilst the cathode/Tank 1 becomes more negatively charged (due to consumption of H^+).

The membrane allows H^+ ions to migrate from Tank 2/anode to Tank 1/cathode so as to maintain electrical neutrality.

- (iv) State and explain what happens to the overall e.m.f when a small amount of $\text{OH}^- (\text{aq})$ ions were added to the electrolyte in **Tank 1**.

[2]

When $\text{OH}^- (\text{aq})$ ions were added, $[\text{H}^+]$ decreases in electrolyte in Tank 1



By Le Chatelier's Principle, the equilibrium position in (1) will shift left to increase $[\text{H}^+]$. Hence, E_{red} becomes less positive

Since, $E_{\text{cell}} = E_{\text{red}} - E_{\text{oxid}}$; E_{cell} becomes less positive/more negative.

- (b) The total concentration of vanadium ions in the 20 dm^3 electrolyte in **Tank 1** is 5.00 mol dm^{-3} . During the charging of the VFB, the percentage composition of VO_2^+ has to be increased from 5% to 90%.

- (i) Calculate the amount of VO_2^+ that needs to be converted during this charging process.

[1]

$$\text{Change in amount of } \text{VO}_2^+ \text{ required} = \left(\frac{90}{100} - \frac{5}{100} \right) \times 20.0 \times 5.00 = 85.0 \text{ mol}$$

- (ii) Find the time taken, in hours, if a 120 A current is used in the charging process.

[2]



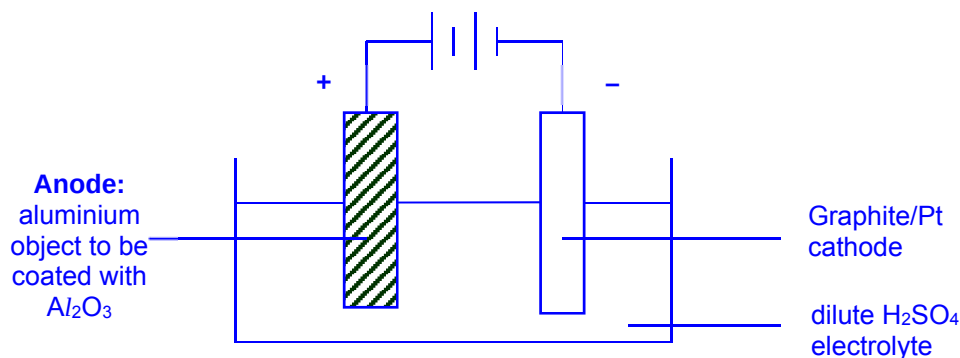
$$\text{Amount of } \text{VO}_2^+ = \frac{It}{nF}$$

$$85.0 = \frac{120 \times t}{1 \times 96500} \quad \rightarrow \quad t = 68354 \text{ s} = 19.0 \text{ h}$$

- (c) Anodising of aluminum is an electrolytic process to increase the thickness of the aluminium oxide layer on the surface of aluminium, by passing a direct current through dilute sulfuric acid.

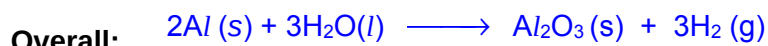
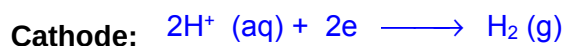
- (i) Draw a labelled diagram of an electrolysis set-up during the anodising of aluminium.

[2]



- (ii) Write the half-equations for the reactions taking place at the anode, cathode and hence write the overall equation.

[2]



- (iii) State and explain any observations if copper(II) sulfate was used as the electrolyte instead.

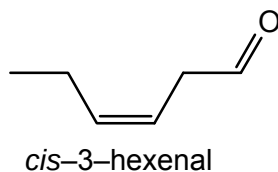
[1]

Pink Cu solid will be deposited on the cathode; blue CuSO_4 solution/electrolyte fades.

Cu^{2+} will be preferentially reduced/discharged at the cathode instead of H^+ , as $E^\ominus_{\text{Cu}^{2+}/\text{Cu}}$ is more positive compared to $E^\ominus_{\text{H}_2/\text{H}^+}$

[Total: 14]

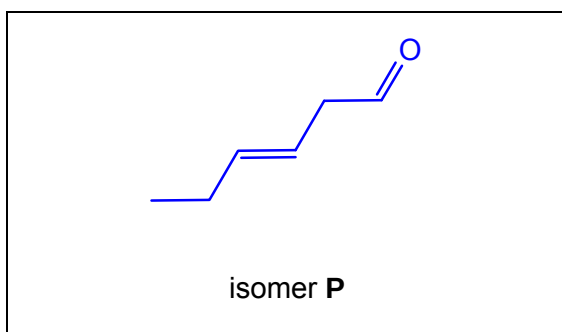
- 4(a) *cis*-3-hexenal, $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CHO}$ is a compound with an intense smell of freshly cut grass and leaves.



Two isomers, **P** and **Q**, with the same molecular formula, $\text{C}_6\text{H}_{10}\text{O}$, as *cis*-3-hexenal show different types of isomerism. Both isomers decolourises aqueous bromine in the dark and both gives a positive test with Fehling's reagent.

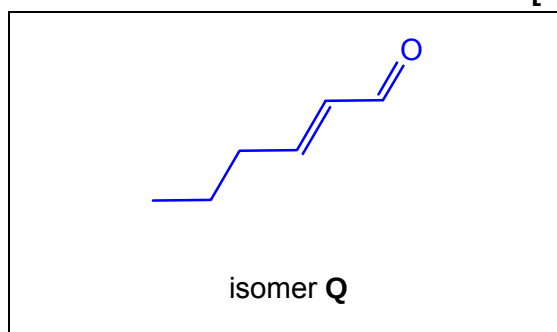
Draw their structures and state their isomeric relationship to *cis*-3-hexenal.

[2]



isomeric relationship to *cis*-3-hexenal:

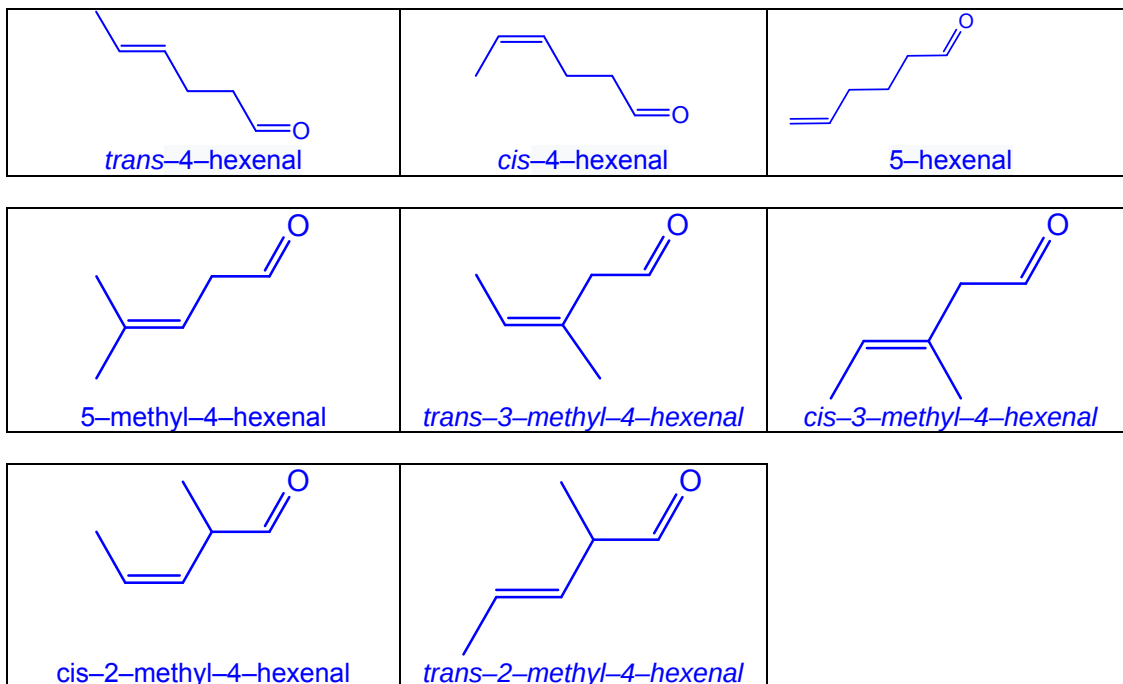
cis-trans isomerism



isomeric relationship to *cis*-3-hexenal:

constitutional isomerism

Other possible answers



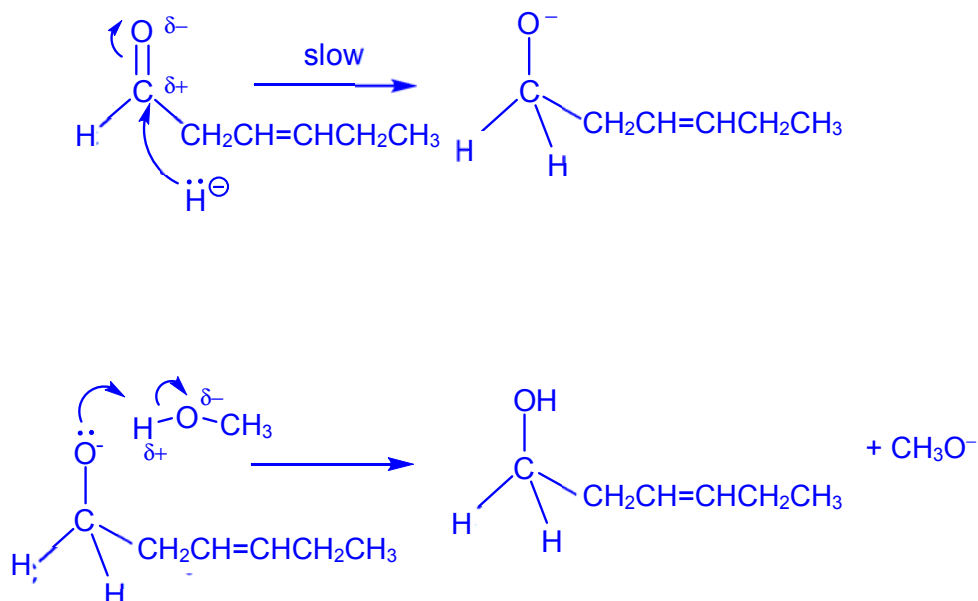
(b) These metal hydride reagents provide a source of hydride, H^- ions.

- (i) The reaction between *cis*-3-hexenal, $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CHO}$ and sodium borohydride in methanol is a nucleophilic addition reaction.

Draw the mechanism for this reaction, showing all charges, dipoles and curly arrows to represent the movement of electron pairs.

[3]

Nucleophilic addition

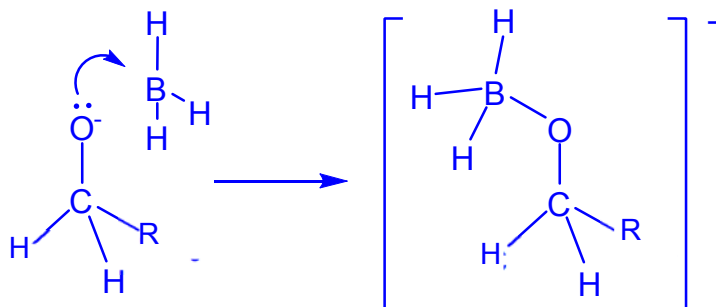


- (ii) BH_3 is a side product of the earlier reaction in (b)(i). BH_3 can form an adduct with Lewis bases, for example, with the alkoxide ion generated in the mechanism.

Draw the structure of this adduct formed and explain why the formation of the adduct is likely to happen.

[2]

- B in BH_3 is electron deficient or B has only 6 electrons.
- B readily accepts lone pair of electron from Lewis bases to form complexes to attain stable octet structure / configuration.



- (c) Both LiAlH_4 and NaBH_4 are sources of H^- . Hydride ions react vigorously with Bronsted–Lowry acids, releasing H_2 (g) and heat.

LiAlH_4 reacts violently with water or alcohols. When carrying out reactions with LiAlH_4 , chemists often use solvents like anhydrous diethyl ether that has been treated to remove traces of water. On the other hand, NaBH_4 does not react that vigorously with water or alcohol.

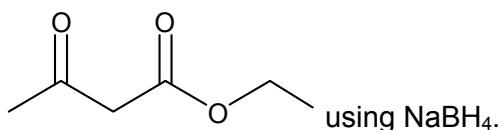
Using the information provided or otherwise, suggest why LiAlH_4 is a better source of H^- than NaBH_4 .

	Pauling Electronegativity
aluminium	1.61
boron	2.04
hydrogen	2.20

[1]

The difference between the electronegativities of B and H is smaller than the difference between the electronegativities of Al and H, so there is a greater degree of ionic character in Al–H bonds than in B–H bonds

- (d) A student proposed the following procedure to reduce ethyl acetoacetate ($\text{C}_6\text{H}_{10}\text{O}_3$),



Procedures

1. Add 0.70 g of NaBH_4 to 25 cm^3 ethanol solvent in a 100 cm^3 round-bottomed flask. Cool the resulting mixture to 0 °C using an ice-bath.
2. To this mixture, slowly add 30 cm^3 of 40.8 g dm^{-3} ethyl acetoacetate solution. Stir the resulting solution at 0 °C for 15 minutes, then allow to warm to room temperature and stir for an additional 15 minutes.
3. Evaporate the solvent. Dissolve the resulting white solid in 30 cm^3 dichloromethane solvent, CH_2Cl_2 .
4. Cool the flask in an ice-bath. Then, add 30 cm^3 of 1 mol dm^{-3} hydrochloric acid dropwise, while stirring, to quench the reaction by destroying any unreacted hydride.
Safety note: The addition of HCl will release H_2 gas.
5. Separate the organic layer from the aqueous layer.
6. Add powdered magnesium sulfate to the organic layer in excess, till no clumping of magnesium sulfate powder is observed. Filter off the magnesium sulfate and evaporate the dichloromethane solvent.
7. Weigh the product and record the actual yield.

- (i) Explain why NaBH_4 is soluble in ethanol (in **step 1**) in terms of structure and bonding. [2]
- NaBH_4 has a giant ionic lattice structure.
 - The formation of ion–dipole interactions between polar ethanol molecules and the ions results in the release of energy causes the detachment of ions from the crystal lattice for solvation/hydration.
 - Hence, NaBH_4 is soluble in ethanol.
- (ii) One mole of NaBH_4 can reduce a maximum of four moles of carbonyl groups. Generally, in an experimental procedure, the reducing agent NaBH_4 should be used in excess.

Determine with relevant calculations, whether the student is correct with the quantities proposed in his procedures (**step 1 – 2**). [2]

$$\text{Amount of NaBH}_4 \text{ available} = \frac{0.70}{23.0 + 10.8 + 4.0} = 0.01852 \text{ mol}$$

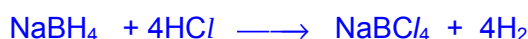
$$\text{Amount of ethyl acetoacetate available} = \frac{40.8 \times \frac{30}{1000}}{12.0 \times 6 + 10.0 + 16.0 \times 3} = 9.415 \times 10^{-3} \text{ mol}$$

Since NaBH_4 can reduced a maximum of 4 carbonyl groups,
mole ratio could be $\text{NaBH}_4 \equiv 4 \text{ carbonyl groups}$

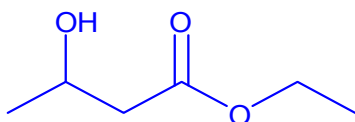
Theoretical amount of NaBH_4 required if all carbonyl groups are used up
 $= 9.415 \times 10^{-3} / 4 = 0.00235 \text{ mol}$

Amount of NaBH_4 available > Amount of NaBH_4 required
 $\Rightarrow \text{NaBH}_4$ is in large excess; the student is correct.

- (iii) In **step 4**, hydrochloric acid is added to quench the reaction, producing hydrogen gas and sodium boron chloride. Write an equation for the reaction that took place. [1]



- (iv) Draw the **skeletal** formula of the product isolated in the organic layer at the end of the experiment. [1]

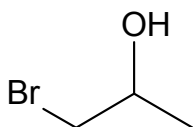


- (v) Suggest the function of magnesium sulfate in **step 6**. [1]
- MgSO_4 : drying agent to remove water in the organic layer

[Total: 15]

- 5 Organic reactions can lead to increased variation due to the direction-specific nature of covalent bonds, giving rise to much complexity from using only a small range of elements such as carbon, hydrogen and oxygen.

- (a) Propene reacts with aqueous bromine to form a mixture of products, including molecule A.



Molecule A

- (i) State the IUPAC name of molecule A.

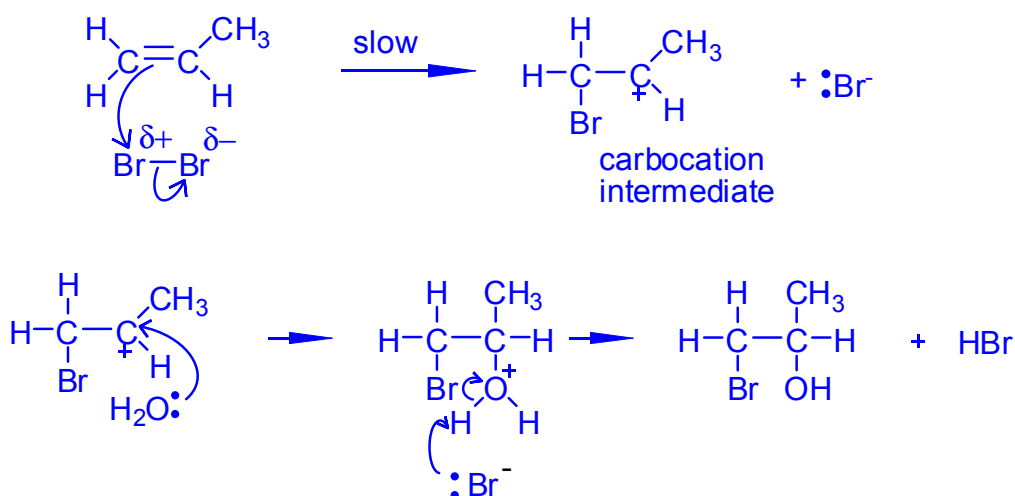
[1]

1-bromopropan-2-ol or 1-bromo-2-propanol

- (ii) Draw a mechanism for the reaction between propene and aqueous bromine to form Molecule A and hydrogen bromide. Show relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs.

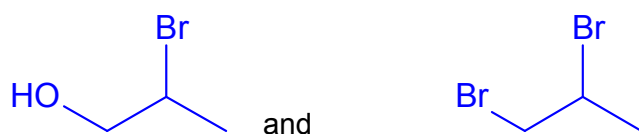
[3]

Electrophilic Addition



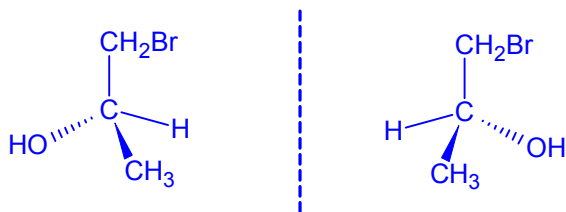
- (iii) Molecule A is the major product of this reaction. Suggest the structure of two other organic by-products.

[2]



- (iv) In the mechanism outlined in (a)(ii), two stereoisomers of **A** are formed in equal amounts. Draw these two stereoisomers and name the type of isomerism displayed.

[2]

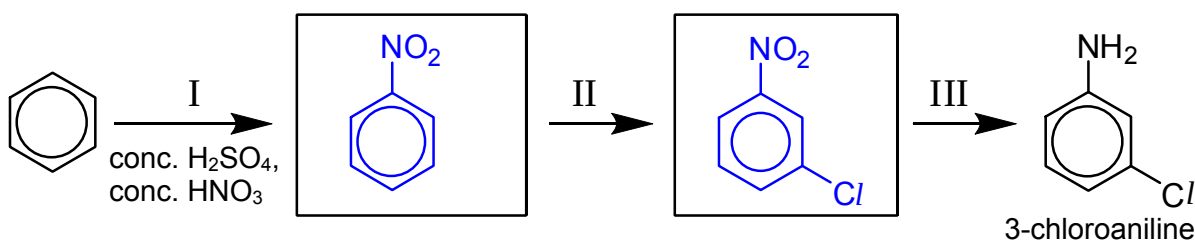


Enantiomerism

- (b) 3-chloroaniline can be synthesised from benzene in a 3-step synthesis.

- (i) Propose a synthetic route by suggesting the two intermediates compounds and state the reagents and conditions used for steps II and III.

[4]



Step II: Cl_2 , anhydrous AlCl_3 , heat

Step III: Sn , in excess conc HCl , heat followed by NaOH (aq)

- (ii) One of the by-products from the synthesis in (b)(i) is HCl , which is a strong acid. The acid dissociation constant values, K_a of several hydrogen halides are shown below.

Compound	$K_a / \text{mol dm}^{-3}$
HCl	1.3×10^6
HBr	1.0×10^9
HI	3.2×10^9

Account for the relative K_a values.

[2]

Down the group, the K_a value of HX increases. This means HX dissociates more readily to give H^+ . This is because H-X bond strength decreases and becomes easier to break.

[Total: 14]